

Basis and Dimension I

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1 Linear combination

1.1 Definition

Definition 2.2.1.

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a subset of V . Then the expression

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m, \text{ with } a_i \in F$$

is called a *linear combination* of S .

1.2 Example

1. $(2, 1, 0)$ is A linear combination of $S = \{(1, 2, 3), (4, 5, 6)\}$ because

$$(2, 1, 0) = -2 \cdot (1, 2, 3) + 1 \cdot (4, 5, 6).$$

2. $f(x) = x^3 + 2x + 1$ in $\mathbb{R}[x]$ is a linear combination of $S = \{1, x, x^2, x^3\}$ because

$$f(x) = 1 \cdot 1 + 2 \cdot x + 0 \cdot x^2 + 1 \cdot x^3.$$

3. **(polynomial)** Show that $\{2x + 1, x^2 + 3\}$ is linearly independent in $\mathbb{R}[x]_{\leq 2}$.

Suppose

$$a_1(2x + 1) + a_2(x^2 + 3) = 0.$$

$$\implies a_2x^2 + 2a_1x + (a_1 + 3a_2) = 0.$$

By comparing coefficients yields

$$a_2 = 0, \quad 2a_1 = 0, \quad a_1 + 3a_2 = 0. \implies a_1 = a_2 = 0$$

Therefore $\{2x + 1, x^2 + 3\}$ is linearly independent.

2 Linear dependence and independence

2.1 Definition

Definition 2.2.2.

1. A **finite** subset $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ of V is called *linearly independent* if

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m = \mathbf{0} \implies a_1 = a_2 = \dots = a_m = 0.$$

In other words, the only way to obtain $\mathbf{0}$ as a linear combination of S is by taking all scalars to 0.

2. More generally, an **infinite** subset S' of V is said to be *linearly independent* if every finite subset $S \subseteq S'$ is *linearly independent*.
3. A subset S'' is called *linearly dependent* if it is *not linearly independent*.

2.2 Example

Prove that, in $\mathbb{R}^3$,

1. $S_1 = \{(1, 2, 3), (4, 5, 6)\}$ is linearly independent:

Proof. Suppose

$$a_1(1, 2, 3) + a_2(4, 5, 6) = (0, 0, 0).$$

Then

$$\begin{cases} a_1 + 4a_2 = 0, \\ 2a_1 + 5a_2 = 0, \\ 3a_1 + 6a_2 = 0. \end{cases}$$

From the first two equations,

$$2(a_1 + 4a_2) - (2a_1 + 5a_2) = 3a_2 = 0.$$

Hence $a_2 = 0$, and therefore $a_1 = 0$, so the trivial solution is the only solution.

$\implies \{(1, 2, 3), (4, 5, 6)\}$ is linearly independent. $\square$

2. $S_2 = \{(1, 2, 3), (4, 5, 6), (7, 8, 9)\}$ is linearly dependent.

Proof. As noted previously,

$$1 \cdot (1, 2, 3) + (-2) \cdot (4, 5, 6) + 1 \cdot (7, 8, 9) = (0, 0, 0)$$

By def, S_2 is not linearly independent. $\square$

2.3 Theorem

Theorem 2.2.1.

$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent iff. either $\mathbf{v}_1 = \mathbf{0}$ or for some r , $\mathbf{v}_r$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$

Proof. ($\Rightarrow$) Given $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is NOT linearly dependent.

Case 1. $\mathbf{v}_1 = \mathbf{0}$. Then there is nothing to prove.

Case 2. $\mathbf{v}_1 \neq \mathbf{0}$.

Since S is linearly dependent, there exist scalars $a_1, \dots, a_m$, not all zero, s.t.

$$a_1 \mathbf{v}_1 + \dots + a_m \mathbf{v}_m = \mathbf{0}.$$

Let r be the largest index such that

$$a_r \neq 0. (a_{r+1} = a_{r+2} = \dots = a_m = 0)$$

Hence (*) becomes

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_r \mathbf{v}_r = \mathbf{0}.$$

Since $a_r \neq 0$, we may solve for $\mathbf{v}_r$:

$$\mathbf{v}_r = -\frac{a_1}{a_r} \mathbf{v}_1 - \frac{a_2}{a_r} \mathbf{v}_2 - \dots - \frac{a_{r-1}}{a_r} \mathbf{v}_{r-1}.$$

Therefore $\mathbf{v}_r$ is a linear combination of

$$\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}.$$

($\Leftarrow$)

Case 1. Suppose $\mathbf{v}_1 = \mathbf{0}$

$$\Rightarrow 1\mathbf{v}_1 + 0\mathbf{v}_2 + 0\mathbf{v}_3 + \dots = 0\mathbf{v}_m = \mathbf{0}$$

$\Rightarrow$ The coefficients are NOT all zero.

Case 2. $\mathbf{v}_1 \neq \mathbf{0}$.

Assume that there $\exists r$ s.t.

$$\mathbf{v}_r = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_{r-1} \mathbf{v}_{r-1}.$$

Then

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_{r-1} \mathbf{v}_{r-1} - \mathbf{v}_r = \mathbf{0}.$$

Since the coefficient of $\mathbf{v}_r$ is $-1 \neq 0$, the coefficients are not all zero.

Therefore there exists a nontrivial linear relation among the vectors in S .

Hence S is linearly dependent.

□

3 Spanning sets

3.1 Definition

Definition 2.3.1. (Linear span)

1. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a finite subset of V . Then define the (linear) span of S by

$$\text{span}(S) := \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m\} : a_i \in F$$

the set of all linear combination of S .

i.e. $\text{span}(S) = \{\text{all linear combination of } S\}$

2. More generally, if S is an infinite subset of V (i.e. S contains infinitely many elements). then we define

$$\text{span}(S) = \bigcup_{S' \subset S} \text{span}(S')$$

where S' runs through all finite subsets of S .

Definition 2.3.2. (Span as a verb)

If $V = \text{span}(S)$ (i.e. every $\mathbf{v} \in V$ can be written as a linear combination of members in S .), we say that S spans V or S is a spanning set of V .

3.2 Remark

By Convention, we have $\text{span}(\emptyset) = \{\mathbf{0}\}$. $\leftarrow$ empty sum is defined to be 0

3.3 Theorem

Theorem 2.3.1.

$\text{span}(S)$ is a subspace of V .

Proof. This is straight-forward, so we do it "verbally":

1. (Closed under addition)

The sum of any two linear combinations of S is still a linear combination of S (by adding up the coefficients).

2. (Closed under scalar multiplication)

A scalar multiple of a linear combination of S is still a linear combination of S (by scaling the coefficients).

3. If S is non-empty, then $\text{span}(S)$ is also non-empty. If S is empty, then

$$\text{span}(S) = \{0\},$$

which is also non-empty.

□

3.4 Example

1.

Let $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} \subseteq \mathbb{R}^3$

$$\begin{aligned}\text{span}(S) &:= \{a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) : a, b, c \in \mathbb{R}\} \\ &= \{(a, b, c) : a, b, c \in \mathbb{R}\} \\ &= \mathbb{R}^3\end{aligned}$$

2.

$S' = \{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$ also spans $\mathbb{R}^3$ because More concretely, for any $(a, b, c) \in \mathbb{R}^3$,

$$\underbrace{(a, b, c) = a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) + 0(1, 1, 1)}_{\text{linear combination of } S'}$$

4 Basis

4.1 Definition

Definition 2.4.1.

A subset S of V is called a *basis* of V if

- (1) S is linearly independent.
- (2) S spans V , i.e. every $\mathbf{v} \in V$ can be written as a linear combination of S .

Moreover, if S can be taken to be a finite set, then we say that V is a *finite-dimensional* vector space. Otherwise, we say that it is *infinite-dimensional*.

Definition 2.4.2.

Fix an order of elements in a basis S . In this case, we call $(a_1, a_2, \dots, a_m)$ the *coordinates* of $\mathbf{v}$ with respect to the basis S and write

$$[\mathbf{v}]_S = (a_1, a_2, \dots, a_m).$$

4.2 Theorem

Theorem 2.4.1.

Suppose

$$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$$

is a basis of V . Then for every $v \in V$, there exists a unique collection of scalars $a_1, a_2, \dots, a_m \in F$ such that

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m.$$

Proof. Since S is a basis of V , it spans V . Therefore every $v \in V$ can be written as a linear combination of the vectors in S .

To prove uniqueness, suppose

$$\mathbf{v} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_m\mathbf{v}_m$$

and

$$\mathbf{v} = b_1\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_m\mathbf{v}_m,$$

where $a_i, b_i \in F$.

Subtracting the two equations, we obtain

$$\mathbf{0} = (a_1 - b_1)\mathbf{v}_1 + (a_2 - b_2)\mathbf{v}_2 + \dots + (a_m - b_m)\mathbf{v}_m.$$

Since S is a basis, it is linearly independent. Hence

$$(a_1 - b_1) = (a_2 - b_2) = \cdots = (a_m - b_m) = 0.$$

Therefore

$$a_1 = b_1, \quad a_2 = b_2, \quad \dots, \quad a_m = b_m.$$

Thus the representation of v is unique. □

4.3 Example

1. Non-uniqueness of a basis

The set

$$S = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$$

is also a basis for $\mathbb{R}^3$ because it is linearly independent and for every $(a, b, c) \in \mathbb{R}^3$,

$$(a, b, c) = (a - b)(1, 0, 0) + (b - c)(1, 1, 0) + c(1, 1, 1).$$

Hence every vector in $\mathbb{R}^3$ is a linear combination of vectors in S .

Moreover, since S is finite, $\mathbb{R}^3$ is finite-dimensional.

2. Infinite-dimensional example

Consider $\mathbb{R}[x]$.

We claim that $\mathbb{R}[x]$ cannot be spanned by any finite set of polynomials.

Proof. Suppose

$$S = \{f_1, f_2, \dots, f_m\}$$

spans $\mathbb{R}[x]$.

Let

$$N = \max\{\deg(f_1), \deg(f_2), \dots, \deg(f_m)\}.$$

Then every polynomial in $\text{span}(S)$ is a linear combination of $f_1, f_2, \dots, f_m$, and therefore has degree at most N .

However,

$$x^{N+1}$$

has degree $N + 1$, so

$$x^{N+1} \notin \text{span}(S).$$

This contradicts the assumption that S spans $\mathbb{R}[x]$.

Therefore, $\mathbb{R}[x]$ is infinite-dimensional. □

3.

As observed previously, for

$$\mathbf{v} = (a, b, c) \in \mathbb{R}^3,$$

we have

- with respect to the standard basis

$$S_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\},$$

we have

$$[\mathbf{v}]_{S_1} = (a, b, c).$$

- (*Order matters*) with respect to the reordered standard basis

$$S'_1 = \{(0, 1, 0), (0, 0, 1), (1, 0, 0)\},$$

we have

$$[\mathbf{v}]_{S'_1} = (b, c, a).$$

- with respect to the basis

$$S_2 = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\},$$

we have

$$[\mathbf{v}]_{S_2} = (a - b, b - c, c).$$

4.3.1 Remark

In particular, the same vector $\mathbf{v}$ may have different coordinates with respect to a different choices of basis.

5 Dimension

5.1 Definition

Definition 2.5.1

Suppose V is a finite-dimensional vector space over F .

Define the dimension of V to be the size of a (and hence any) basis of V :

$$\dim_F(V) = |S|$$

where S is a basis of V .

5.2 Theorem

Theorem 2.5.1.(Exchange Theorem / Replacement Theorem)

Suppose

- $S_1 = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a spanning set of V ;
- $S_2 = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ is linearly independent.

Then

$$m \leq n.$$

Proof. This proof will be provided after developing *sifting method* theorem in other chapter. □

Theorem 2.5.2. (Basis Theorem)

Suppose V is a finite-dimensional vector space over F . If S_1 and S_2 are both bases of V , then S_1 and S_2 contain the same number of elements.

Proof. Let S_1 and S_2 be two bases of V .

- Since S_1 is a basis, it is linearly independent.

Since S_2 is a basis, it is a spanning set.

By the Exchange Theorem,

$$|S_1| \leq |S_2|.$$

- Since S_2 is a basis, it is linearly independent.

Since S_1 is a basis, it is a spanning set.

By the Exchange Theorem,

$$|S_2| \leq |S_1|.$$

Hence,

$$|S_1| = |S_2|.$$

Therefore every basis of V has the same cardinality. □

5.3 Example

1.

Since

$$S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

is a basis of $\mathbb{R}^3$ (as a vector space over $\mathbb{R}$), we conclude that

$$\dim_{\mathbb{R}}(\mathbb{R}^3) = |S| = 3.$$